

4.16 Revised ERD

Second Submission 2/14/26 by Steve Gonzales for IMT543

In this evolution of the ERD, I introduced user management system, making the schema multi-tenant capable rather than single-user focused. The addition of budget, financial_transaction, expense_details, income_details, and savings_goal tables transforms the system from simple transaction logging to comprehensive financial planning.

Steps taken:

- **Generalization.** Looking at the income and expenses, I identified commonalties like user and amount. For example, the amount for an expense is typically negative and the amount for income is positive. When we generalize this data into a Supertype, we can then quickly add those amounts together with a simple query.
- **Supertype/Subtype.** Instead of separate transaction, income and expense tables, I have created a financial_transaction table (supertype) and then subtype tables for income_details and expense_details. This allows us to quickly do financial reporting across both categories, income and expense instead of having to write multiple queries and union them or do some other operation to combine them.
- **Specialization.** Looking at the expense and income tables in the previous model, I tried to identify attributes that only make sense in each table separately. For example, income:
 - Doesn't have a "payment method" (it's received, not paid)
 - Doesn't have a "merchant" (it comes from employer/source)
 - Doesn't have "receipts" in the same way.

So then we include these fields in the expense_details and exclude from income_details. This provides us with the following benefits:

- Domain Accuracy: Each type has only relevant attributes
- Data Integrity: Constraints specific to each type
- Query Efficiency: Don't load irrelevant data
- Maintainability: Changes to one type don't affect others
- Extensibility: Easy to add new specializations

This evolution employs enhanced data modeling patterns like:

- Generalization and Specialization
- Proper normalization: Better separation of concerns
- Constraint-driven design: Business rules enforced at database level
- Scalable architecture: Supports complex financial scenarios

The revised ERD for my Project includes these added tables:

- `Savings Goal`. This allows the user to enter financial goals that can then track progress. E.g. if you were saving to put a down payment on a house.
- `Budget`. This allows the user to have set and follow a spending budget. It will include total monthly amounts and actuals can then be tracked against the budget.
- `Financial_transaction` A combined table that includes both income and expenses. This generalized table allows us to quickly and efficiently aggregate financial data .
- `Expense_details`. This subtype rolls up to the transaction table, but will only include expenses for the purpose of tracking the budget.
- `Income_details`. This subtype also rolls up to the transaction table. A specialized table that will track income coming in. This subtype rolls up to the `financial_transaction` table.

category
123 category_id
A-Z category
A-Z cat_sub
A-Z cat_super

merchant
123 merchant_id
A-Z merchant_name
A-Z merchant_type

institution
123 institution_id
A-Z institution_name

expense_details
123 transaction_id
A-Z category
A-Z payment_method
123 category_id
123 account_id
123 merchant_id
☒ is_recurring
A-Z receipt_url

account
123 account_id
A-Z account_name
123 institution_id
A-Z account_number

income_details
123 transaction_id
123 account_id
A-Z source
A-Z income_type
☒ is_taxable
A-Z employer_name
A-Z payment_frequency
123 gross_amount
123 net_amount

budget
123 budget_id
123 user_id
A-Z category_name
123 monthly_limit
123 current_spent
123 budget_year
123 budget_month

financial_transaction
123 transaction_id
123 user_id
123 amount
A-Z description
🕒 transaction_date
A-Z transaction_type

savings_goal
123 goal_id
123 user_id
A-Z goal_name
123 target_amount
123 current_amount
🕒 target_date
123 priority_level
☒ is_active
🕒 created_at

users
123 user_id
A-Z username
A-Z email
A-Z first_name
A-Z last_name
🕒 created_at